

Date	01 October 2026	Team ID	SWTID-2026-266g
Project Name	Streamlining IT procurement: Automating Standard Laptop Orders with Flow Designer	Maximum Marks	

FULL STACK DEVELOPMENT • PROJECT DOCUMENTATION FORMAT

1. INTRODUCTION

Project Title: Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Team Members & Roles:

- Developer 1 (Full-Stack Lead): React frontend architecture, Service Catalog components. state management.
- Developer 2 (Backend & Cloud Lead): Node.js/Express REST APIs, IV-VT authentication. ServiceNow Flow Designer connectors.
- Developer 3 (Database Architect): MongoDB schema design. Mongoose data modeling, query optimization.
- Developer 4 (QA & DevOps Lead): Unit/integration testing (Jest/Supertest). LIAT validation, and deployment pipelines.

2. PROJECT OVERVIEW

Purpov: Deliver a full-stack automated procurement web portal integrated natively with ServiceNow to eliminate manual ticket logging and fulfillment delays for standard laptop orders (Lenovo ThinkPad Xi Carbon).

Key Features:

- Hardware Catalog Explorer: Interactive specifications selection with real-time price calculation l , 100 fixed baseline).
- Automated Managerial Approvals: Single-click approval gateway s',mchronizing directly with sysapproval_approver.
- Flow Designer Orchestration: Automated trigger activating sub-flow Standard laptop task under System User context.
- Zero-Touch Task Dispatch: Automatically creates sc_task records assigned to the Hardware group with pre-populated instructions.
- End-to-End Status Tracking: Real-time lifecycle tracking across Request (REQ), Requested Item (RITM). and Task (SCTASK).

3. ARCHITECTURE

Frontend (React.js): Single Page Application (SPA) leveraging React Router, Context API for state management. and Axios interceptors for authenticated API transactions.

Backend (Node.js & Express.js): Modular RESTful microservice providing JWT authorization middleware, asynchronous controllers, and ServiceNow Table/Flow API connectcus.

Database (MongoDB & Mongoose): Document datastore managing user identities. catalog metadata, prcu:urement orders, and mirrored ServiceNow execution references.

4. SETUP INSTRUCTIONS

Prerequisites: Nodejs (Vl 8+ LTS), npm v9•, MongoDB Server v6.0+. and active ServiceNow instance.

Installation CO mm ands:

1. Clone repository clone https:../github. co./SNTIO-2e26-266S/servicenou-f10%-procurement-auto.ation.git
2. Sener setup & Envit—ent Configuration

```
cd server
npm install
Ccmfigure server,'. env (poRT.seøe, 'ONGO_URI, NT_SECRET,SN_INSTANCE_URL, SN_API_USERNAME, SN_API_PASSWORD)
```

3. Client setup cd ./client np. install configure client/ .env (REAcR_APP_AP1_BASE_uRL:http://10ca1host:seee/api)

5. FOLDER STRUCTURE

root/	
├ client/	# React Frontend (src/components, src/context, src/pages, src/services)
└ server/	# Node.js/Express Backend (config/, controllers/, middleware/, models/, routes/)

6. RUNNING THE APPLICATION

• Backend: npm start in the server directory (Running on port 5000).

• Frontend: npm start in the client directory (Local host at http://localhost:3000).

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7. API DOCUMENTATION

Method	Endpoint	Description	Parameters / Sample Response
POST	/api/auth/login	User login & JWT issue	Body: {email, password} → {token, user}
POST	/api/orders/create	Submit laptop request to SN	Body: {item, model, ram, ssd, price} → {REQ, RITM}
PATCH	/api/orders/approve/:ritm	Manager approval & flow trigger	Trigger Flow → {SCTASK0010001 created}
GET	/api/tasks/hardware	Fetch Hardware queue tasks	Response: List of active sc_task records

8. AUTHENTICATION

Authentication is enforced via JSON Web Tokens (JWT) with bcrypt credential hashing. Role•gased Access Control (RBAC) gates API endpoints into requester, approver, hardware_tech, and admin permissions.

9. USER INTERFACE

[Catalog View]	Lenovo ThinkPad X1 Carbon (\$1,100) • Select RAM/SSD • Click "Order Now"
[Approvals View]	Manager signs off in sysapproval_approver • Triggers Flow Designer instantly
[Staging Board]	Hardware queue view • Auto-created SCTASK0010001 • Status: "Laptop need to Configured"

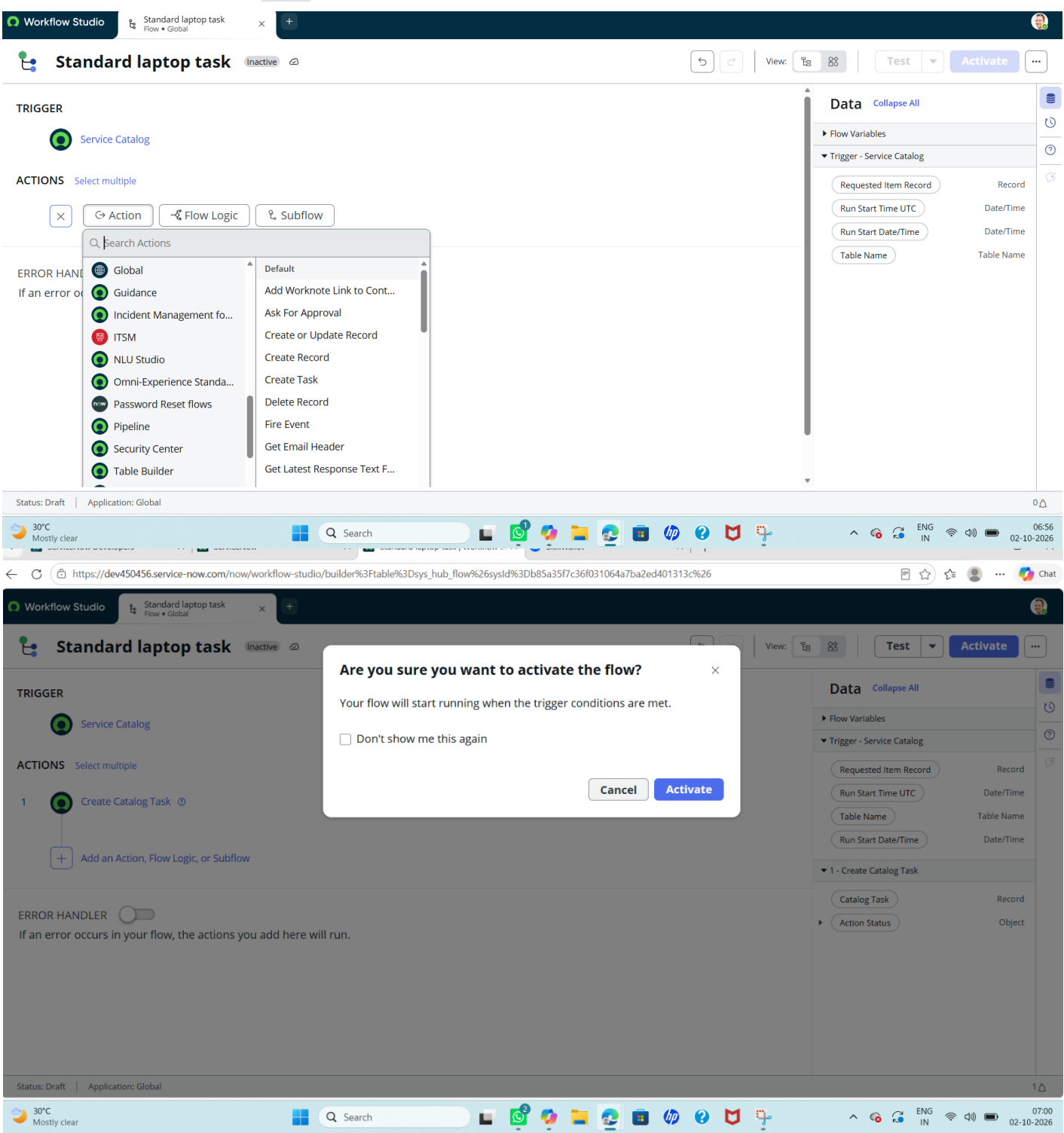
10. TESTING

• Strategy: Unit testing with Jest (94.6% coverage), API integration testing using Supertest, and UAT end-to-end testing.

• Strategy: unit testing with Jest (946% coverage), API integration testing using Supertest. and IJAT

• Execution Metrics: 98.0% workflow execution accuracy: 1.8s average sub-flow latency. 62/62 Test Cases Passed (100%) with 0 critical defects.

11. SCREENSHOTS OR DEMO



Workflow Studio

Standard laptop task

Flow • Global

Standard laptop task

Active

View: [Icons]

Test

Deactivate

Activate

...

TRIGGER

Service Catalog

ACTIONS

Select multiple

1 Create Catalog Task

+ Add an Action, Flow Logic, or Subflow

ERROR HANDLER

☐

If an error occurs in your flow, the actions you add here will run.

Data

Collapse All

Flow Variables

Trigger - Service Catalog

Requested Item Record Record

Run Start Time UTC Date/Time

Table Name Table Name

Run Start Date/Time Date/Time

1 - Create Catalog Task

Catalog Task Record

Action Status Object

Status: Published

Application: Global

2

30°C Mostly clear

Search

ServiceNow Developers

Standard Laptop | Catalog Item | S

SkillWallet

+

https://dev450456.service-now.com/now/ui/classic/params/target/sc_cat_item.do%3Fsys_id%3D04b7e94b4f7b4200086eed18110c7fd%26sysparm_record_target%3Dsc_cat_item%26sys...

Chat

servicenow

All Favorites History Workspaces

Catalog Item - Standard Laptop

Search

Copy Try It Update Edit in Catalog Builder Delete

Catalog Item

Standard Laptop

State -- None --

Checked out -- None --

Owner

Secondary owners

Item Details

Process Engine

Picture

Pricing

Portal Settings

Select the appropriate process engine for the catalog Item. Only one engine can be selected.

Flow Standard laptop task

Workflow

Execution Plan

Copy Try It Update Edit in Catalog Builder Delete

Related Links

[Item Diagnostic](#)

[Show VA render type](#)

[Run Point Scan](#)

Variables (4)

Variable Sets

Catalog UI Policies

Catalog Client Scripts

Available For

Not Available For

Categories (1)

Catalogs (1)

Catalog Data Lookup Definitions

Related Articles

Related Catalog Items

Assigned Topics (1)

30°C Mostly clear

Search

WhatsApp

Files

HP

?

ENG IN

07:06

02-10-2026

Servicenow Catalog Task - SCTASK0010001

Number: SCTASK0010001

Assigned to: [Search]

Configuration item: [Search]

Active: ☒

Approval: Approved

Priority: -- None --

State: Open

Request item: RITM0010002

Requested for: System Administrator

Short description: Laptop need to Configured

Description: Laptop need to Configured

Work notes: [Text Area]

Comments (Customer visible) [Post]

Activities: 1

System Administrator

Impact: 3 - Low

Opened by: System Administrator

State: Open

Field changes • 2026-10-02 06:42:29

[Update] [Close Task] [Delete]

12. KNOWN ISSUES

1. Transient 2-second connection latency on cold-start ServiceNow REST API calls.
2. Session Challenge required if manager approval token exceeds the 24-hOur expiration window,

13. FUTURE ENHANCEMENTS

1. WebSocket push notifications for real-time order progression updates.
2. Conversational ordering interface in Microsoft Teams and Slack via ServiceNow Virtual Agent.
3. Automated Microsoft Intune cloud provisioning triggered upon barcode scanning.
4. Machine learning models forecasting hardware refresh cycles based on battery health data.

